

ROBUST OPTIMIZATION IN ACTIVE LINEAR REGRESSION

Khai Dong, Dylan Le, & Kyle Eng

Department of Computer Science
Rensselaer Polytechnic Institute
Troy, NY 12180, USA

ABSTRACT

This paper considers the problem of reducing label complexity for large datasets, as a means to reduce the cost of performing supervised learning with fewer labels. Despite the strengths of machine learning applications, training for high-dimensional problems requires greatly more labeled data to achieve optimum regression weights. As these learning algorithms become larger in dimension, more data will be needed. The human cost of labeling is great, so it is desired to reduce the need for labeling data as much as possible. In this paper we present our chaining sampling algorithm that reduces the label complexity of a dataset. We examine two case studies to see the effectiveness of this algorithm in real and synthetic data.

1 INTRODUCTION

In many real-world domains, we can collect huge amounts of unlabeled data (millions of records, images, signals), but labels are expensive and slow as they require experts (e.g., radiologists, pathologists, clinicians). As such, we want to minimize the number of labels needed while still enjoying the statistical benefits of having a large dataset. Let $X \in \mathbb{R}^{N \times d}$ and $y \in \mathbb{R}^N$ be the design matrix and target vector, respectively. The optimal regression weights w_* minimize $\|Xw - y\|_2^2$ over $w \in \mathbb{R}^d$. The goal of our paper is to query a small number of labels in y without knowing y , and produce a set of near-optimal weights w' using only the revealed labels and the corresponding rows in X . Specifically, we desire w' to be a $(1 + d/N)$ -approximation to w_* i.e.,

$$\|Xw' - y\|_2^2 \leq (1 + d/N)\|Xw_* - y\|_2^2 \quad (1)$$

since under typical modeling assumptions, the expected squared prediction error of w_* approaches the Bayes optimal error to within $O(d/N)$ (Abu-Mostafa et al., 2012, Chapter 3).

1.1 PRIOR WORKS

Research on solving this problem has been particularly active, driven both by theoretical interest and by practical applications in areas such as machine learning, signal processing, and numerical linear algebra, etc., Chkifa et al. (2015); Cohn et al. (1994); Guo et al. (2020); Konyushkova et al. (2015); Luo et al. (2004). Multiple algorithms for active learning have been proposed Beygelzimer et al. (2009); Boutsidis et al. (2013); Castro & Nowak (2008); Chen & Price (2019); Dasgupta (2011); Dasgupta & Hsu (2008); Dereziński & Warmuth (2017); Dereziński et al. (2018); Drineas et al. (2008); Gittens & Magdon-Ismail (2023); Kremer et al. (2014); Sabato & Munos (2014), most of which operate by sampling rows of X to be included in the coreset. The most common algorithm is leverage score sampling, which achieves an ϵ -regret with a coreset of size $O(d/\epsilon + d \log d)$ Drineas et al. (2008); Mahoney (2011); Woodruff (2014). Several other algorithms, such as leveraged volume sampling Dereziński et al. (2018) and pivotal dependent sampling Shimizu et al. (2024), asymptotically achieve the same coreset size. Then, Chen & Price (2019) and Devvrit et al. (2022) give 2 algorithms that achieve a coreset of size $O(d/\epsilon)$; these algorithms are based on the randomized BSS proposed in Lee & Sun (2015). However, at $\epsilon = d/N$, pivotal dependent sampling and randomized BSS provide very little label complexity reduction (either due to the large multiple constant or the structure of algorithm itself), while leverage score sampling and leverage

volume sampling can only yield $(1 - e^{-2}) \cdot N$ label complexity reduction in the worst case Dong & Magdon-Ismail (2026). Then, Gittens & Magdon-Ismail (2023) proposed an algorithm to reject rows specifically for $\epsilon = d/N$; the algorithm reduces label complexity by $\mathcal{O}(\sqrt{N})$. These algorithms, while they can't achieve the same label complexity reduction as their counterparts, exhibit superior performance for the removal of 1 data points. In this work, we wish to expand on Gittens & Magdon-Ismail (2023)'s work.

2 NOTATIONS

Firstly, $X \in \mathbb{R}^{N \times d}$ and $y \in \mathbb{R}^N$ are the target design matrix and the corresponding labels of X , respectively. Typically, we assume $\text{poly}(d) \ll N \ll e^d$ and X is of full rank. $x_i \in \mathbb{R}^d$ are the columns of X^T and y_i is the corresponding label of x_i . Denote X_A^- the matrix where we remove rows in A from X . Similarly, y_A^- and y_A are the labels of X_A^- and A , respectively. For the sake of simplicity, we assumed for all considered A , X_A^- is at full rank. Denote w_* the regression weight obtained by regressing X with y and w_A^- the regression weight obtained by regressing X_A^- with y_A^- . Let $X_A^- = U\Sigma V^T$ be the thin SVD. Then, $u_{A,i} \in \mathbb{R}^d$ is the i^{th} column of U^T . For brevity, we denote $u_{\emptyset,i} = u_i$. Let $X = U\Sigma V^T$ be the thin SVD. Then, U_A and U_A^- are the rows of U corresponding with A and X_A^- , respectively, and $P_A = U_A U_A^T$ is the partial projection matrix.

3 PRELIMINARY LEMMAS

We first proved some useful lemmas.

Lemma 1. *Gittens & Magdon-Ismail (2023)[Equation (17)] If $I - A$ is invertible, then $(I - A)^{-1}A = A(I - A)^{-1} = (I - A)^{-1} - I$.*

Lemma 2. *Let $X \in \mathbb{R}^{N \times d}$ be of full rank. Let $X = U\Sigma V^T$ be the thin SVD of X . Then, for all A, B comprised of rows of X , $U_A U_B^T = A(X^T X)^{-1} B^T$ where U_A, U_B are comprised of rows of U corresponding to rows of A and B in X .*

Proof. By considering the SVD of X , we have the following

$$A = U_A \Sigma V^T, \quad B = U_B \Sigma V^T, \quad \text{and} \quad (X^T X)^{-1} = V \Sigma^{-2} V^T. \quad (2)$$

Thus,

$$A(X^T X)^{-1} B^T = U_A \Sigma V^T \cdot V \Sigma^{-2} V^T \cdot V \Sigma U_B^T = U_A U_B^T, \quad (3)$$

as desired. $\blacksquare$

Lemma 3.

$$\|u_{A,i}\|_2^2 = \|u_i\|_2^2 + u_i^T U_A^T (I - P_A)^{-1} U_A u_i. \quad (4)$$

Proof. Indeed,

$$\|u_{A,i}\|_2^2 = x_i^T (X^T X - A^T A)^{-1} x_i \quad (5)$$

$$= x_i^T [(X^T X)^{-1} + (X^T X)^{-1} A^T (I - P_A)^{-1} A (X^T X)^{-1}] x_i \quad (6)$$

$$= \|u_i\|_2^2 + u_i^T U_A^T (I - P_A)^{-1} U_A u_i \quad (7)$$

where (7) follows from Lemma 2, as desired. $\blacksquare$

Lemma 4. *Define $\alpha_{A,i} = A((X_A^-)^T X_A^-)^{-1} x_i$. Then,*

$$\alpha_{A,i} = (I - P_A)^{-1} U_A u_i \quad \text{and} \quad \|\alpha_{A,i}\|_2^2 = u_i^T U_A^T (I - P_A)^{-2} U_A u_i. \quad (8)$$

Proof. By the Woodbury matrix inversion identity, we have that

$$\alpha_{A,i} = A(X^T X - A^T A)^{-1} x_i \quad (9)$$

$$= A \left[(X^T X)^{-1} + (X^T X)^{-1} A^T (I - A(X^T X)^{-1} A^T)^{-1} A(X^T X)^{-1} \right] x_i \quad (10)$$

$$= U_A u_i + P_A (I - P_A)^{-1} U_A u_i \quad (11)$$

$$= (I - P_A)^{-1} U_A u_i \quad (12)$$

where (11) and (12) follow from Lemma 2 and Lemma 1, respectively, as desired. The rest of the lemma follows by evaluating $\|\alpha_{A,i}\|_2^2$. ■

Lemma 5.

$$\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2 = \|u_i\|_2^2 + u_i^T U_A^T [(I - P_A)^{-1} + (I - P_A)^{-2}] U_A u_i \quad (13)$$

Proof. Follows from Lemma 3 and Lemma 4. ■

Lemma 6. Let $U \in \mathbb{R}^{N \times d}$, $D \in \mathbb{R}^{N \times N}$ be a diagonal matrix, and $y \in \mathbb{R}^N$. Then,

$$U^T D y = \sum_{i=1}^N D_{ii} u_i y_i. \quad (14)$$

Proof. This is trivial from multiplying out $U^T D y$. ■

4 CHAINING SAMPLING ALGORITHM

Prior work Gittens & Magdon-Ismail (2023) showed an approximation ratio of $1 + \mathcal{O}(d/N^2)$ by removing 1 data point from X . It also lays out the arguments for the following conjecture. This means we can keep removing 1 data point while preserving an approximation ratio of $1 + \mathcal{O}(d/N^2)$ from the previous dataset, we could potentially remove $\Omega(N)$ data points from X .

Theorem 7. (Gittens & Magdon-Ismail, 2023, Theorem 2.2) For any X , pick row i to throw out with probability $p_i \propto \frac{(1-\ell_i^2)}{\ell_i}$. Then,

$$\mathbb{E} [\|X w_i^- - y\|^2] \leq \left(1 + \frac{d}{(N-d)^2} \right) \|X w_* - y\|^2. \quad (15)$$

Conjecture 1. Pick any X and pick out k rows $A_k = \{x_1, x_2, \dots, x_k\}$ to throw out with some probability distribution $\mathbb{P}(x_i | A_{i-1})$. Then,

$$\mathbb{E} [\|X w_{A_k}^- - y\|^2] \leq \left(\prod_{i=0}^{k-1} \left(1 + \frac{d}{(n-d-i)^2} \right) \right) \|X w_* - y\|^2. \quad (16)$$

In this work, we will refine this approach and demonstrate that, under the following assumptions on X , a slightly weaker result than Conjecture 1 can still be obtained.

Algorithm 1 Chaining Sampling Algorithm

Ensure: $\mathbb{E}[\|X w_A^- - y\|_2^2] \leq (1 + d/N) \|X w_* - y\|_2^2$

1: $A = \emptyset$

2: $\gamma = 0$

3: **while** $1 + \gamma \leq 1 + d/N$ **do**

4: Sampling rows x_i from X_A^- with some probability $p_{A,i}$

5: $A = A \cup \{x_i\}$.

6: Recompute the approximation ratio γ

7: **return** X_A^- .

▷ removed data.

▷ expected approximation ratio

Lemma 8. Let A be k rows of X and x_i be a row from X_A^- . Let $A' = A \cup \{x_i\}$. Then,

$$\begin{aligned} \|Xw_{A'}^- - y\|_2^2 &= \|Xw_A^- - y\|_2^2 + \frac{\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2}{(1 - \|u_{A,i}\|_2^2)^2} \|x_i^T w_A^- - y_i\|_2^2 \\ &\quad + 2(1 - \|u_{A,i}\|_2^2)^{-1} (Aw_A^- - y_A)^T \alpha_{A,i} (x_i^T w_A^- - y_i). \end{aligned} \quad (17)$$

Proof. Using the identities $w_* = (X^T X)^{-1} X^T y$, $(X_A^-)^T X_A^- = (X_{A'}^-)^T X_{A'}^- + x_i x_i^T$, $(X_A^-)^T y_A^- = (X_{A'}^-)^T y_{A'}^- + x_i y_i$, and applying the Woodbury matrix inversion identity, we have that

$$\begin{aligned} w_{A'}^- &= ((X_A^-)^T X_A^- - x_i x_i^T)^{-1} \cdot ((X_A^-)^T y_A^- - x_i y_i) \end{aligned} \quad (18)$$

$$= \left[((X_A^-)^T X_A^-)^{-1} + ((X_A^-)^T X_A^-)^{-1} x_i (1 - \|u_{A,i}\|_2^2)^{-1} x_i^T ((X_A^-)^T X_A^-)^{-1} \right] \cdot ((X_A^-)^T y_A^- - x_i y_i) \quad (19)$$

$$= w_A^- + ((X_A^-)^T X_A^-)^{-1} x_i (1 - \|u_{A,i}\|_2^2)^{-1} (x_i^T w_A^- - y_i), \quad (20)$$

where (20) follows from Lemma 1. Consider $\|Xw_{A'}^- - y\|_2^2$,

$$\begin{aligned} \|Xw_{A'}^- - y\|_2^2 &= \|Xw_A^- - y\|_2^2 \\ &\quad + (1 - \|u_{A,i}\|_2^2)^{-2} \|x_i^T w_A^- - y_i\|_2^2 \|X((X_A^-)^T X_A^-)^{-1} x_i\|_2^2 \\ &\quad + 2(Xw_A^- - y)^T X((X_A^-)^T X_A^-)^{-1} x_i (1 - \|u_{A,i}\|_2^2)^{-1} (x_i^T w_A^- - y_i). \end{aligned} \quad (21)$$

With some rigorous algebra, we can prove that

$$\|X((X_A^-)^T X_A^-)^{-1} x_i\|_2^2 = \|\alpha_{A,i}\|_2^2 + \|u_{A,i}\|_2^2. \quad (22)$$

Thus,

$$\begin{aligned} \|Xw_{A'}^- - y\|_2^2 &= \|Xw_A^- - y\|_2^2 + \frac{\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2}{(1 - \|u_{A,i}\|_2^2)^2} \|x_i^T w_A^- - y_i\|_2^2 \\ &\quad + 2(1 - \|u_{A,i}\|_2^2)^{-1} (Aw_A^- - y)^T \alpha_{A,i} (x_i^T w_A^- - y_i), \end{aligned} \quad (23)$$

since $(X_A^- w_A^- - y_A^-)$ is orthogonal to X_A^- . ■

Theorem 9. Let A be a set of k rows in X . Sample a row x_i in X_A^- with $p_{A,i} \propto \frac{(1 - \|u_{A,i}\|_2^2)^2}{\|u_{A,i}\|_2^2}$. Let $A' = A \cup \{x_i\}$. Then,

$$\mathbb{E} [\|Xw_{A'}^- - y\|^2] \leq \|Xw_A^- - y\|^2 + \frac{d \cdot (1 + \kappa_A)}{(n - d - k)^2} \|X_A^- w_A^- - y_A^-\|^2 + \mathcal{E}_A \quad (24)$$

where $\kappa_A = \|P_A\|_2 (2 - \|P_A\|_2) (1 - \|P_A\|_2)^{-2}$ and

$$|\mathcal{E}_A| \leq \mu_A \cdot \frac{d}{(n - d - k)^2} \cdot \frac{\|P_A\|_2^{1/2}}{1 - \|P_A\|_2} \cdot \|Aw_A^- - y_A\| \|X_A^- w_A^- - y_A^-\| \quad (25)$$

where $\mu_A = \text{range} \left\{ \frac{1}{\|u_{A,i}\|_2^2} \right\}_{x_i \in X_A^-}$.

Proof. By Lemma 8, we have

$$\begin{aligned} \|Xw_{A'}^- - y\|_2^2 &= \|Xw_A^- - y\|_2^2 + \frac{\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2}{(1 - \|u_{A,i}\|_2^2)^2} \|x_i^T w_A^- - y_i\|_2^2 \\ &\quad + 2(1 - \|u_{A,i}\|_2^2)^{-1} (Aw_A^- - y_A)^T \alpha_{A,i} (x_i^T w_A^- - y_i). \end{aligned} \quad (26)$$

Thus, by Lemma 4,

$$\begin{aligned} \mathbb{E} [\|Xw_A^- - y\|_2^2] &= \|Xw_A^- - y\|_2^2 + \frac{1}{Z_A} \sum_i \frac{\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2}{\|u_{A,i}\|_2^2} \|x_i^T w_A^- - y_i\|_2^2 \\ &\quad + \frac{2}{Z_A} (Aw_A^- - y_A)^T (I - P_A)^{-1} U_A \sum_i \frac{1 - \|u_{A,i}\|_2^2}{\|u_{A,i}\|_2^2} u_i (x_i^T w_A^- - y_i). \end{aligned} \quad (27)$$

By Lemma 3, Lemma 5, and (Golub & Van Loan, 1996, Lemma 2.3.3), we have that

$$\frac{\|u_{A,i}\|_2^2 + \|\alpha_{A,i}\|_2^2}{\|u_{A,i}\|_2^2} = \frac{\|u_i\|_2^2 + u_i^T U_A^T [(I - P_A)^{-1} + (I - P_A)^{-2}] U_A u_i}{\|u_i\|_2^2 + u_i^T U_A^T (I - P_A)^{-1} U_A u_i} \quad (28)$$

$$\leq 1 + \|P_A\|_2 [(1 - \|P_A\|_2)^{-1} + (1 - \|P_A\|_2)^{-2}] \quad (29)$$

$$= 1 + \|P_A\|_2 (2 - \|P_A\|_2) (1 - \|P_A\|_2)^{-2}. \quad (30)$$

For brevity, define $\kappa_A = \|P_A\|_2 (2 - \|P_A\|_2) (1 - \|P_A\|_2)^{-2}$. Then, by Lemma 6, we have that

$$\sum_i \frac{1 - \|u_{A,i}\|_2^2}{\|u_{A,i}\|_2^2} u_i (x_i^T w_A^- - y_i) = (U_A^-)^T D_1 (X_A^- w_A^- - y_A^-). \quad (31)$$

for $D_1 = \text{Diag} \left(\left[\frac{1 - \|u_{A,i}\|_2^2}{\|u_{A,i}\|_2^2} \right]_i \right)$. Putting (29), (31), and Lemma 2 into (27), we have that

$$\begin{aligned} \mathbb{E} [\|Xw_A^- - y\|_2^2] &\leq \|Xw_A^- - y\|_2^2 \\ &\quad + \frac{1 + \kappa_A}{Z_A} \|X_A^- w_A^- - y_A^-\|_2^2 \\ &\quad + \frac{2}{Z_A} (Aw_A^- - y_A)^T (I - P_A)^{-1} A (X^T X)^{-1} (X_A^-)^T D_1 (X_A^- w_A^- - y_A^-). \end{aligned} \quad (32)$$

We observe that $(X_A^-)^T (X_A^- w_A^- - y_A^-) = 0$. Thus, we can subtract any multiple of I from D_1 in (32). Thus,

$$\begin{aligned} \mathbb{E} [\|Xw_A^- - y\|_2^2] &\leq \|Xw_A^- - y\|_2^2 \\ &\quad + \frac{1 + \kappa_A}{Z_A} \|X_A^- w_A^- - y_A^-\|_2^2 \\ &\quad + \frac{2}{Z_A} (Aw_A^- - y_A)^T (I - P_A)^{-1} U_A (U_A^-)^T D_2 (X_A^- w_A^- - y_A^-) \end{aligned} \quad (33)$$

where

$$D_2 = \text{Diag} \left(\left[\frac{1}{\|u_{A,i}\|_2^2} \right]_i \right) - \frac{1}{2} \left[\left(\min_i \frac{1}{\|u_{A,i}\|_2^2} \right) + \left(\max_i \frac{1}{\|u_{A,i}\|_2^2} \right) \right] I. \quad (34)$$

Define $\mu_A = \max_i \frac{1}{\|u_{A,i}\|_2^2} - \min_i \frac{1}{\|u_{A,i}\|_2^2}$. By construction, $\|D_2\|_2 \leq \mu_A/2$. Denote

$$\mathcal{E}_A = \frac{2}{Z_A} (Aw_A^- - y_A)^T (I - P_A)^{-1} U_A (U_A^-)^T D_2 (X_A^- w_A^- - y_A^-). \quad (35)$$

Then,

$$|\mathcal{E}_A| \leq \frac{\mu_A}{Z_A} \cdot \frac{\|P_A\|_2^{1/2}}{1 - \|P_A\|_2} \cdot \|Aw_A^- - y_A\|_2 \|X_A^- w_A^- - y_A^-\|_2. \quad (36)$$

Put everything together, we have that

$$\mathbb{E} [\|Xw_A^- - y\|_2^2] \leq \|Xw_A^- - y\|_2^2 + \frac{1 + \kappa_A}{Z_A} \|X_A^- w_A^- - y_A^-\|_2^2 + \mathcal{E}_A, \quad (37)$$

By the same argument in (Gittens & Magdon-Ismail, 2023, Theorem 1), $Z_A \geq \frac{(N-k-d)^2}{d}$, as desired. ■

Under Assumption 1, κ_A is bounded by a constant. In Algorithm 1, at every iteration, we can compute $\|P_A\|_2$ in recomputing approximation ratio $1 + \gamma$ to maximize the amount of data removed. Then, by Assumption 2, $\mu_A \in \mathcal{O}(d/N)$. Thus, we can remove $\Omega(N)$ data points from X if either by Assumption 2 or Assumption 3 holds.

Assumption 1. For $|A| \leq \lfloor N/2 \rfloor$, $\|P_A\|_2 \leq 1/2$.

Assumption 2. For $\max_i \|u_{A,i}\|_2^2 - \min_i \|u_{A,i}\|_2^2 \in \mathcal{O}(d/N^2)$.

Assumption 3. $\sum_i \mathcal{E}_{A_i} \in \mathcal{O}(d/N)$.

In practice, since A is known at runtime, we can just stop when Assumption 1 does not hold. Assumption 2, however, does not hold for a lot of X . However, in most cases, the algorithm succeed in reducing the label complexity by $\mathcal{O}(N)$; what was observed is Assumption 3 as $\mathcal{E}_A$ can be less than 0, which allows the terms to cancel itself out.

5 EXPERIMENTS

In this section, we evaluate the Chaining Sampling Algorithm (CSA) on both synthetic and real datasets, with the goal of approximating regression weights using fewer observed labels.

5.1 SETUP

The primary objective is to determine how many labels we can safely "discard" while maintaining a predictive error close to the full-dataset solution. We evaluate the performance across varying dimensions d and sample sizes N . In particular, benchmark CSA against some baseline sampling methods and validate the approximation ratio $(1 + d/N)$.

We consider the synthetic and real setting in our experiments. In the synthetic setting we create two examples using scikit-learn parametric datasets (Pedregosa et al., 2011). In the first example we create a randomized regression problem with $d = 10$ with 5 informative features sampled with Gaussian noise with standard deviation 0.002. In the second example we use the classical two moons dataset for classification with standard deviation noise of 0.005. In both examples we use $N = 1000$ samples. In the real-world datasets we consider the wine quality dataset from Cortez & Reis (2009) and the concrete compressive strength dataset from Yeh (1998). We measure the relationship of the approximation ratio to the label budget. In each example we remove up-to 60% of the labels to produce our experiments. We further just do naive linear regression without further feature engineering.

5.2 RESULTS

We discuss what assumptions are broken by our datasets that make CSA fail.

In Figure 1, we initialize a dataset with $N = 1000$ samples in $d = 10$ dimensions, along with randomly generated labels. Applying CSA, we are able to remove up to 50% of the labels while remaining within the target approximation bound.

As shown in Figure 1, the approximation error decreases steadily as the label budget increases, and remains consistently below the theoretical threshold of d/N . Notably, CSA maintains a stable approximation ratio even after discarding a significant portion of the dataset. This result indicates that many data points contribute little to the overall regression fit. The smooth decay in error suggests that the CSA algorithm effectively targets uninformative samples in the dataset, leading to a stable decrease in error over a large range of the label budget.

These results are consistent with the theoretical guarantees, as the observed approximation error remains stable and does not exhibit exponential decay that might be expected from iterative data removal. This result provides empirical support for the effectiveness of the CSA algorithm.

To further explore this behavior, we consider additional synthetic benchmarks. In Figure 2a, we evaluate CSA on the two-moons dataset. In this setting, CSA is only able to remove approximately 10% of the labels before reaching the target approximation bound, representing a substantial deviation from the conjectured 50% removal. This suggests that when the data contains strong non-linearity, a larger fraction of samples becomes essential for preserving an accurate fit.

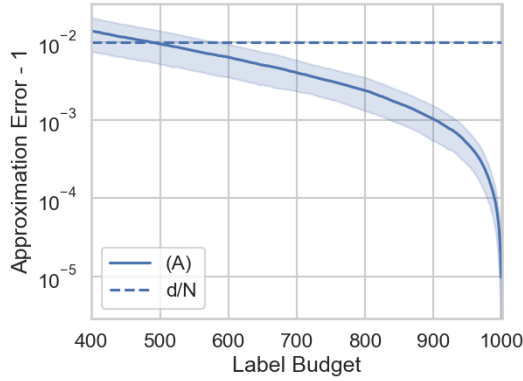


Figure 1: CSA can easily have removed points if noise is random rather than adversarial. (A) is CSA. The shaded region is within 1-standard deviation.

In contrast, Figure 2b shows results on a synthetic regression benchmark with less restrictive structure and more uniform contributions from each instance. CSA is able to remove close to 50% of the labels while remaining within the target approximation bound, closely aligning with the conjectured behavior observed in earlier experiments.

This motivates further investigation into more realistic settings, where data distributions may be more structured or exhibit complex dependencies. Evaluating CSA on real data will reveal more about capabilities and limitations of CSA in practice.

We evaluate CSA on several benchmark regression datasets commonly used in empirical studies. Specifically, we consider the wine quality datasets, considering both the red and white wine variants, and the concrete compressive strength dataset. Unlike the synthetic setting, these datasets reflect the nontrivial interactions among features that are characteristic of real-world data, which violate the assumptions under which CSA performs optimally.

For each dataset, we apply CSA under the same experimental protocol as in the synthetic case, varying the label budget and measuring the resulting approximation error relative to the full dataset solution. Figure 3 summarizes the performance of CSA across different label budgets for each dataset.

Applying CSA to both the red and white wine quality datasets, we observe in Figure 3a and 3b, we are able to remove up to approximately 30% of the labels while remaining within the target approximation bound.

For the concrete compressive strength dataset, we observe in Figure 3c that CSA is able to remove a larger percentage of the labels before reaching the target approximation bound, up to nearly 50%.

These results highlight how deviations from idealized assumptions impact the performance of CSA. In particular, the presence of real world data, which exhibits complex feature interactions and variable sample importance, limits CSA’s ability to reliably identify and discard uninformative points, leading to degraded approximation quality.

Overall, the experiments across both synthetic and real-world datasets reveal a mixed alignment with the conjectured behavior of CSA. In some settings, particularly those where the data exhibits a uniform sample importance, CSA achieves performance close to the theoretical expectation, resulting in near 50% label removal. However, in cases where data exhibits more structure or non-linearity, in both synthetic and real-world data, CSA deviates from the conjecture and exhibits significantly reduced label removal capacity.

6 CONCLUSION AND DISCUSSION

We have observed that CSA is very powerful when the model is well specified, but it performs poorly in the presence of deterministic noise. However, all tested real datasets yield roughly 10% –

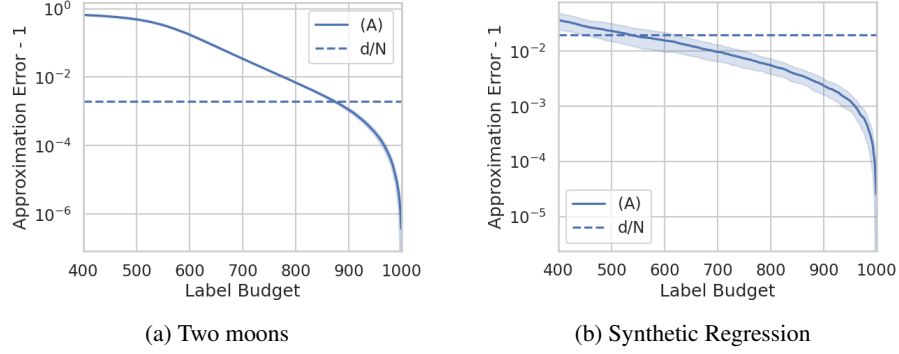


Figure 2: Synthetic datasets. In 2a, we only remove roughly 10% of the rows before we reach the approximation ratio, this is likely due to being a classification problem. In 2b, we can remove nearly 50% of the labels in the dataset matching our conjecture.

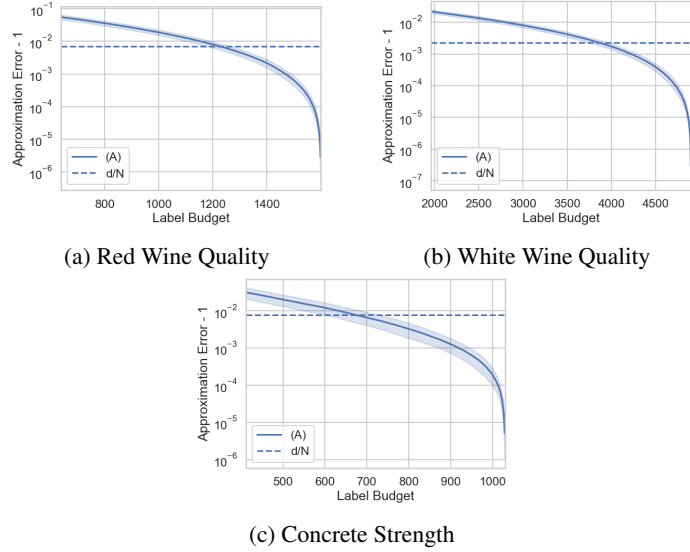


Figure 3: We remove roughly 30% of the labels before our approximation ratio is achieved in the wine datasets. The concrete dataset removes roughly 50%.

30% label reduction, which is a substantial cost reduction as the number of samples N increases. However, due to the lack of a concrete coreset size, this algorithm is yet to be useful in practice, as we do not have access to y for verification.

It is an open question whether we can determine the label reduction only given X , if our sampling distribution is optimal, and whether we can find a distribution in which we may achieve the same label reduction under Gaussian noise, theoretically and empirically, under adversarial noise.

REFERENCES

- Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin. *Learning From Data: A Short Course*. AMLBook, 2012. ISBN 1600490069.
- Alina Beygelzimer, Sanjoy Dasgupta, and John Langford. Importance weighted active learning. In *Proceedings of the 26th Annual International Conference on Machine Learning, ICML '09*, pp. 49–56, New York, NY, USA, 2009. Association for Computing Machinery. ISBN 9781605585161. doi: 10.1145/1553374.1553381. URL <https://doi.org/10.1145/1553374.1553381>.
- Christos Boutsidis, Petros Drineas, and Malik Magdon-Ismail. Near-optimal coresets for least-squares regression. *IEEE Transactions on Information Theory*, 59(10):6880–6892, 2013. doi: 10.1109/TIT.2013.2272457.
- R. M. Castro and R. D. Nowak. Minimax bounds for active learning. *IEEE Trans. Inf. Theor.*, 54(5):2339–2353, May 2008. ISSN 0018-9448. doi: 10.1109/TIT.2008.920189. URL <https://doi.org/10.1109/TIT.2008.920189>.
- Xue Chen and Eric Price. Active regression via linear-sample sparsification. In Alina Beygelzimer and Daniel Hsu (eds.), *Proceedings of the Thirty-Second Conference on Learning Theory*, volume 99 of *Proceedings of Machine Learning Research*, pp. 663–695. PMLR, 6 2019. URL <https://proceedings.mlr.press/v99/chen19a.html>.
- Abdellah Chkifa, Albert Cohen, Giovanni Migliorati, Fabio Nobile, and Raul Tempone. Discrete least squares polynomial approximation with random evaluations - application to parametric and stochastic elliptic pdes. *ESAIM: M2AN*, 49(3), 2015. doi: 10.1051/m2an/2014050.
- David Cohn, Les Atlas, and Richard Ladner. Improving generalization with active learning. *Mach. Learn.*, 15(2):201–221, May 1994. ISSN 0885-6125. doi: 10.1023/A:1022673506211. URL <https://doi.org/10.1023/A:1022673506211>.
- Cerdeira A. Almeida F. Matos T. Cortez, Paulo and J. Reis. Wine Quality. UCI Machine Learning Repository, 2009. DOI: <https://doi.org/10.24432/C56S3T>.
- Sanjoy Dasgupta. Two faces of active learning. *Theor. Comput. Sci.*, 412(19):1767–1781, April 2011. ISSN 0304-3975. doi: 10.1016/j.tcs.2010.12.054. URL <https://doi.org/10.1016/j.tcs.2010.12.054>.
- Sanjoy Dasgupta and Daniel Hsu. Hierarchical sampling for active learning. In *Proceedings of the 25th International Conference on Machine Learning, ICML '08*, pp. 208–215, New York, NY, USA, 2008. Association for Computing Machinery. ISBN 9781605582054. doi: 10.1145/1390156.1390183. URL <https://doi.org/10.1145/1390156.1390183>.
- Michał Dereziński and Manfred K. Warmuth. Unbiased estimates for linear regression via volume sampling. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*, pp. 3087–3096, Red Hook, NY, USA, 2017. Curran Associates Inc. ISBN 9781510860964.
- Michał Dereziński, Manfred K. Warmuth, and Daniel Hsu. Leveraged volume sampling for linear regression. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems, NIPS'18*, pp. 2510–2519, Red Hook, NY, USA, 2018. Curran Associates Inc.

- Fnu Devvrit, Nived Rajaraman, and Pranjal Awasthi. Semi-supervised active linear regression. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS '22*, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.
- Khai Dong and Malik Magdon-Ismail. Review: Oblivious coresets for tight active linear regression. arXiv, 2026.
- Petros Drineas, Michael W. Mahoney, and S. Muthukrishnan. Relative-error cur matrix decompositions. *SIAM Journal on Matrix Analysis and Applications*, 30(2):844–881, 2008. doi: 10.1137/07070471X. URL <https://doi.org/10.1137/07070471X>.
- Alex Gittens and Malik Magdon-Ismail. Reduced label complexity for tight ℓ_2 regression, 2023. URL <https://arxiv.org/abs/2305.07486>.
- G.H. Golub and C.F. Van Loan. *Matrix Computations*. Johns Hopkins Studies in the Mathematical Sciences. Johns Hopkins University Press, 1996. ISBN 9780801854149. URL <https://books.google.com/books?id=m1Oa7wPX6OYC>.
- Ling Guo, Akil Narayan, and Tao Zhou. Constructing least-squares polynomial approximations. *SIAM Review*, 62(2):483–508, 2020. doi: 10.1137/18M1234151. URL <https://doi.org/10.1137/18M1234151>.
- Ksenia Konyushkova, Raphael Sznitman, and Pascal Fua. Introducing geometry in active learning for image segmentation. In *Proceedings of the 2015 IEEE International Conference on Computer Vision (ICCV), ICCV '15*, pp. 2974–2982, USA, 2015. IEEE Computer Society. ISBN 9781467383912. doi: 10.1109/ICCV.2015.340. URL <https://doi.org/10.1109/ICCV.2015.340>.
- Jan Kremer, Kim Steenstrup Pedersen, and Christian Igel. Active learning with support vector machines. *Wiley Int. Rev. Data Min. and Knowl. Disc.*, 4(4):313–326, July 2014. doi: 10.1002/widm.1132. URL <https://doi.org/10.1002/widm.1132>.
- Yin Tat Lee and He Sun. Constructing linear-sized spectral sparsification in almost-linear time. In *2015 IEEE 56th Annual Symposium on Foundations of Computer Science*, pp. 250–269, 2015. doi: 10.1109/FOCS.2015.24.
- Tong Luo, Kurt Kramer, Dmitry B. Goldgof, Lawrence O. Hall, Scott Samson, Andrew Remsen, and Thomas Hopkins. Active learning to recognize multiple types of plankton. In *Proceedings of the Pattern Recognition, 17th International Conference on (ICPR'04) Volume 3 - Volume 03, ICPR '04*, pp. 478–481, USA, 2004. IEEE Computer Society. ISBN 0769521282.
- Michael W. Mahoney. Randomized algorithms for matrices and data. *Found. Trends Mach. Learn.*, 3(2):123–224, February 2011. ISSN 1935-8237. doi: 10.1561/22000000035. URL <https://doi.org/10.1561/22000000035>.
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- Sivan Sabato and Remi Munos. Active regression by stratification. In *Proceedings of the 28th International Conference on Neural Information Processing Systems - Volume 1, NIPS'14*, pp. 469–477, Cambridge, MA, USA, 2014. MIT Press.
- Atsushi Shimizu, Xiaouu Cheng, Christopher Musco, and Jonathan Weare. Improved active learning via dependent leverage score sampling. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=IYxDy2jDFL>.
- David P. Woodruff. Sketching as a tool for numerical linear algebra. *Found. Trends Theor. Comput. Sci.*, 10(1–2):1–157, October 2014. ISSN 1551-305X. doi: 10.1561/04000000060. URL <https://doi.org/10.1561/04000000060>.
- I-Cheng Yeh. Concrete Compressive Strength. UCI Machine Learning Repository, 1998. DOI: <https://doi.org/10.24432/C5PK67>.